DEPARTMENT 1 · DEVELOPMENT AND IT · PROFESSION 10

3D modeller

Profession 10 of 10 in Development and IT.

What the profession means

A 3D modeller creates three-dimensional objects and scenes for product visualisation, animation, games, architecture and advertising.

What professionals do every day

Models objects from briefs and reference, sets materials and lighting, renders images, checks scale and topology, and exports for the client's pipeline.

Why businesses hire them, and what problems they solve

Products can be shown before they exist, from every angle, in any colour. 3D reduces photo shoots and speeds approvals.

- Objects with wrong scale or messy geometry.
- Renders with poor lighting that do not look real.
- Files that cannot be used in the client's software.

Skills you need

- Modelling basics: vertices, edges, faces.
- Materials, textures and UV mapping.
- Lighting and rendering.
- File formats (glTF, OBJ, FBX).

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
Blender	3D modelling, animation and rendering	Free	Free	www.blender.org
Blender manual	Official reference	Free	Free	docs.blender.org/manual/en/latest/
SketchUp	Architecture and product modelling	Free web version	Paid plans	www.sketchup.com
Sketchfab	Share and view 3D models	Free tier	Paid plans	sketchfab.com

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish
Beginner	Model a simple product (a mug) to correct scale.
Intermediate	Texture it and light it for a product image.

Advanced	Export a glTF file and check it in a viewer.

Projects by level

- **Beginner:** Product visualisation for a shop.
- **Intermediate:** Architectural interior render.
- **Advanced:** Animated product spin.

Client-ready portfolio project

A product visualisation for a fictional bottle brand: model to scale, materials, lighting, three render views and a glTF export with a file note.

Worked example

Worked example. A 500 ml bottle is modelled at 500 ml scale, so the render matches the real product in the client's catalogue.

Assignment with full solution

ASSIGNMENT

Scaled model and export

Business problem: A brand needs a bottle model for its catalogue at true scale.

Steps

1. Model the bottle using the given dimensions.
2. Apply a glass material and a label texture.
3. Light with a three-point setup and render three views.
4. Export glTF and check it opens in a web viewer.

Expected output: Renders, a glTF file and a note on units and scale.

SOLUTION (ANSWER KEY)

Units are set at the start so dimensions are true. The glTF export carries the materials; the viewer check confirms the file works outside Blender.

Common mistakes

- Ignoring scale until the end.
- Overly dense meshes that slow the viewer.
- Exporting without checking textures.
- Lighting that is too flat.

Freelancing opportunities

- Product renders for online shops.

- Interior and architectural visualisation.
- Briefs for 'Blender', '3D product model', '3D rendering'.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- Blender manual (official)
- Blender tutorials (official)

Part of **Master Freelancing: Design and Creative**. Back to the chapter list on skillnavora.in · Open the full textbook. Official links in this chapter were not checked live; confirm before relying on them.